

Experimental Results

1. Original Images



(a)



(b)



(c)



(d)



(e)



(f)

Fig 1 Original test images used for evaluation: (a) Airplane (b) Barbara (c) House (d) Lena (e) Cameraman and (f) Peppers.

2. Embedded Images



(a)



(b)



(c)



(d)



(e)



(f)

Fig 2 Embedded Images of the original pictures: (a) airplane (b) barbara (c) house (d) lena (e) cameraman and (f) peppers

3. Tamper Detection and Recovery Results

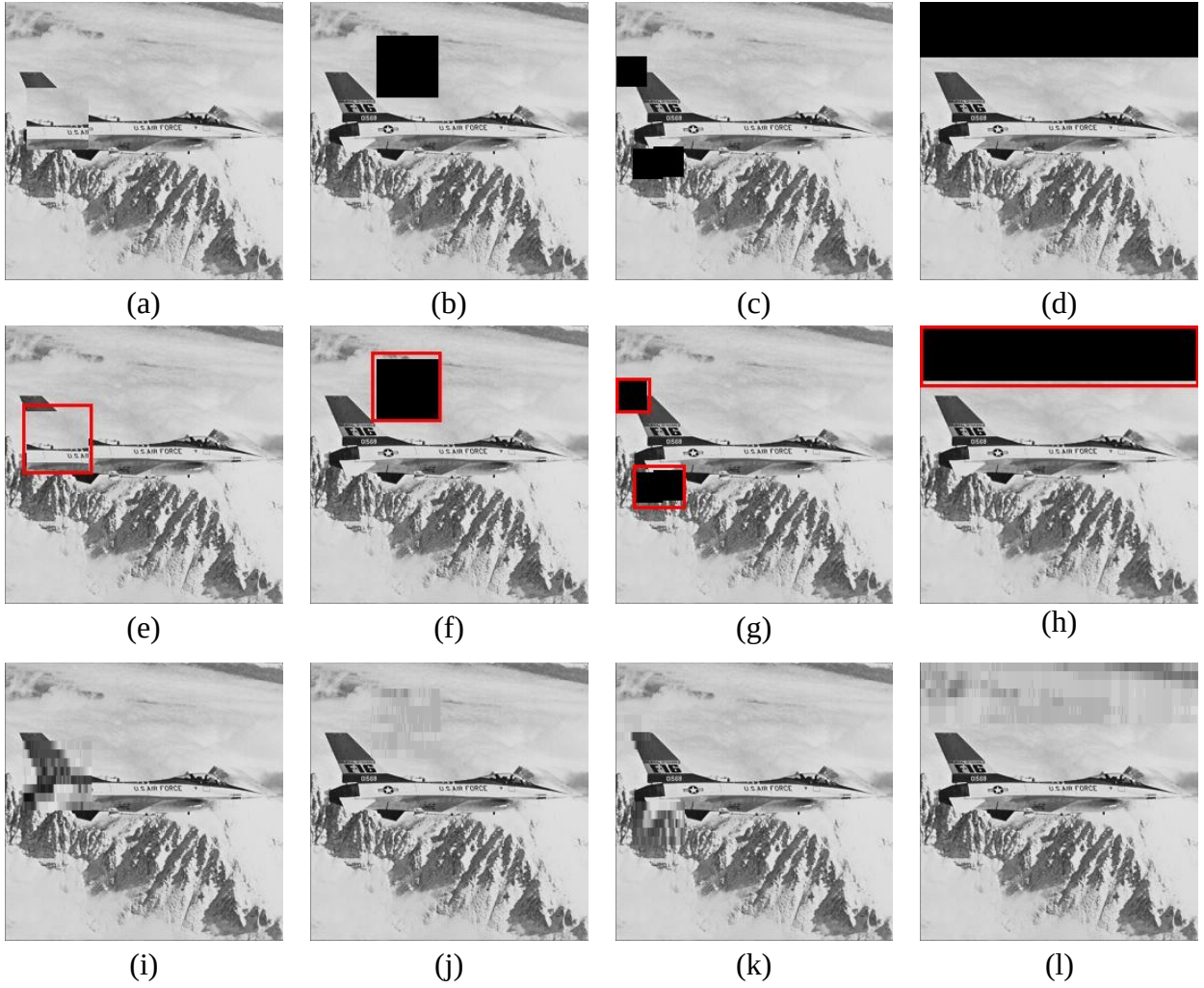


Fig 3 Tampering and recovery results: (a) 5% copy-move attack, (b) 5% cropping attack, (c) multiple cropping attack (5%), and (d) top-strip removal attack (20%). **(e)–(h) show the corresponding images for (a)–(d) with bounding boxes applied to highlight the tampered regions, respectively.** (i)–(l) show the corresponding recovered images for (a)–(d), respectively.

4. Performance Evaluation

Performance analysis of tampering and recovery on Airplane, Barbara, pepper images using PSNR, MSE, and SSIM

Image			Original vs Tampered			Original vs Recovered		
	Attack name	Attack percent	PSNR	MSE	SSIM	PSNR	MSE	SSIM
Airplane	Multiple crop	1.37%	20.03	645.2	0.967	38.39	9.404	0.985
	Multiple crop	5%	18.95	826.6	0.95	26.56	143.2	0.95
	Multiple Crop	10%	12.71	34807	0.87	28.04	101.9	0.94
	Crop	1.37%	21.71	438.21	0.977	32.99	32.63	0.981
	Crop	5%	15.34	1900	0.94	37.7	10.8	0.97
	Crop	10%	13.46	2925.6	0.88	26.94	131.4	0.94
	Bottom-Strip	20%	9.59	7141.6	0.79	28.48	92.08	0.90
	Top-Strip	20%	9.4	7344.5	0.793	31.09	50.57	0.91
	Copy Paste	1.37%	31.3	47.44	0.98	33.19	31.17	0.984
	Copy Paste	10%	19.53	723.43	0.89	25.79	171.04	0.93
	Copy Paste	5%	20.92	525.09	0.946	27.85	106.6	0.96
Barbara	Multiple crop	1.37%	25.7	171.3	0.98	36.25	15.38	0.98
	Multiple crop	5%	18.15	993.9	0.93	34.01	25.8	0.97
	Multiple Crop	10%	16.6	1402.1	0.87	32.7	34.7	0.96
	Crop	1.37%	29.6	69.9	0.9	40.1	6.2	0.99
	Crop	5%	20.3	605	0.93	37.5	11.5	0.97
	Crop	10%	17.3	1207.3	0.89	34.1	24.8	0.96
	Bottom-Strip	20%	15.31	1912.5	0.79	28.78	86.05	0.92
	Top-Strip	20%	13.91	2640.5	0.798	27.82	107.26	0.921
	Copy Paste	1.37%	33.05	32.15	0.98	41.9	4.18	0.98
	Copy Paste	10%	19.8	671.2	0.9	32.1	39.5	0.94
	Copy Paste	5%	24.42	234.8	0.94	31.4	46.9	0.95
Pepper	Multiple crop	1.37%	25.4	184.09	0.97	36.14	15.79	0.98
	Multiple crop	5%	20.3	602.6	0.93	34.4	23.09	0.97
	Multiple Crop	10%	14.5	2265.2	0.87	30.14	62.89	0.94
	Crop	1.37%	22.9	333.4	0.97	40.8	5.3	0.99
	Crop	5%	17.7	1102.3	0.93	31.3	47.9	0.96
	Crop	10%	15.63	1775.5	0.88	30.5	57.5	0.95
	Bottom-Strip	20%	12.9	3320.5	0.79	26.81	135.3	0.90

	Top-Strip	20%	12.4	3689.6	0.79	27.69	110.54	0.91
	Copy Paste	1.37%	29.7	68.5	0.98	39.4	7.3	0.98
	Copy Paste	10%	21.08	506.9	0.9	32.9	33.1	0.95
	Copy Paste	5%	21.5	456.9	0.94	31.1	49.8	0.96

5. Original and Synthetic Images

ORIGINAL IMAGE



SYNTHETIC IMAGE

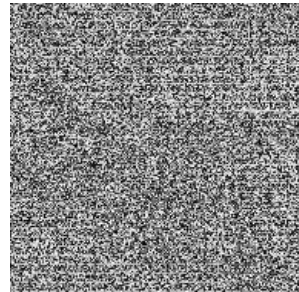


Fig 4 Original Image and its corresponding Synthetic image of Hash and Recovery Bits with the size of 192×192

6. AWS S3 Cloud Storage

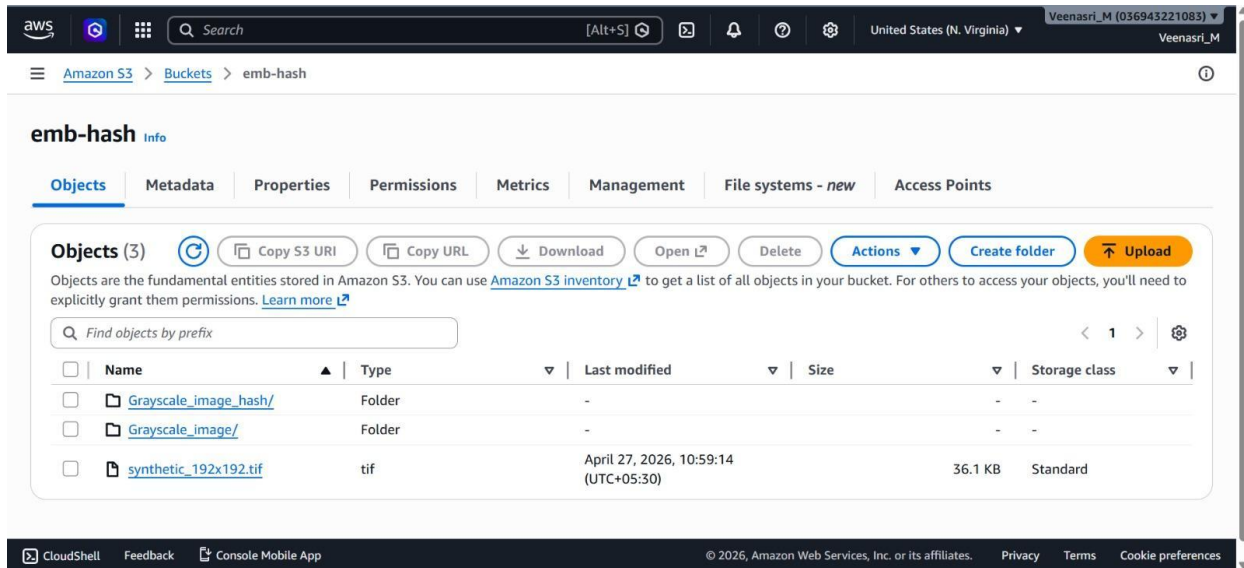


Fig 5 Amazon S3 cloud storage used in the proposed framework

Figure 5 illustrates the use of Amazon S3 cloud storage as a communication channel in the proposed tamper detection and recovery framework. The sender uploads the synthetic image along with its associated hash data to the cloud, which is then accessed by the receiver for integrity verification and recovery. This approach ensures reliable and efficient data transfer between system components.